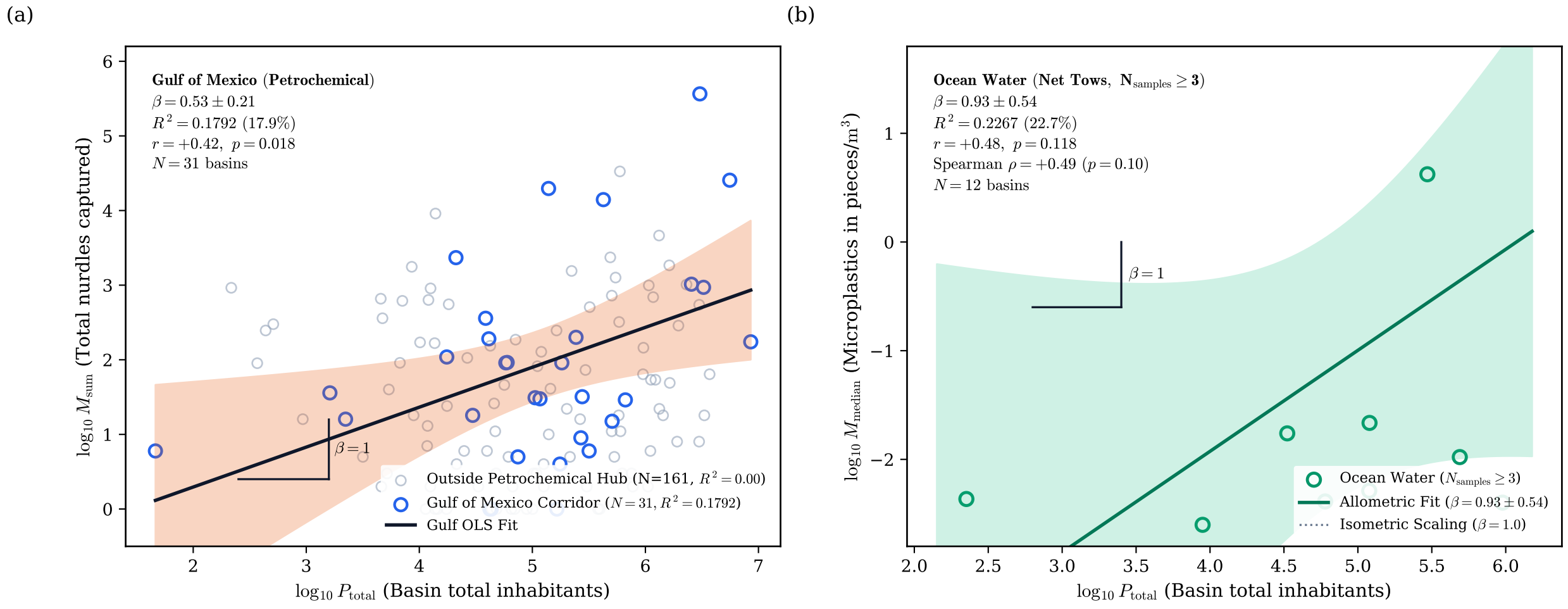

(a)
Gulf of Mexico (Petrochemical)
$\beta = 0.53 \pm 0.21$
$R^2 = 0.1792$ (17.9%)
$r = +0.42$, $p = 0.018$
$N = 31$ basins
$\beta = 1$
Outside Petrochemical Hub (N=161, $R^2 = 0.00$)
Gulf of Mexico Corridor ($N = 31$, $R^2 = 0.1792$)
Gulf OLS Fit
$\log_{10} M_{\text{sum}}$ (Total nurdles captured)
$\log_{10} P_{\text{total}}$ (Basin total inhabitants)
(b)
Ocean Water (Net Tows, $\mathbf{N_{samples} \geq 3}$)
$\beta = 0.93 \pm 0.54$
$R^2 = 0.2267$ (22.7%)
$r = +0.48$, $p = 0.118$
Spearman $\rho = +0.49$ ($p = 0.10$)
$N = 12$ basins
$\beta = 1$
Ocean Water ($N_{\text{samples}} \geq 3$)
Allometric Fit ($\beta = 0.93 \pm 0.54$)
Isometric Scaling ($\beta = 1.0$)
$\log_{10} M_{\text{median}}$ (Microplastics in pieces/m^3)
$\log_{10} P_{\text{total}}$ (Basin total inhabitants)